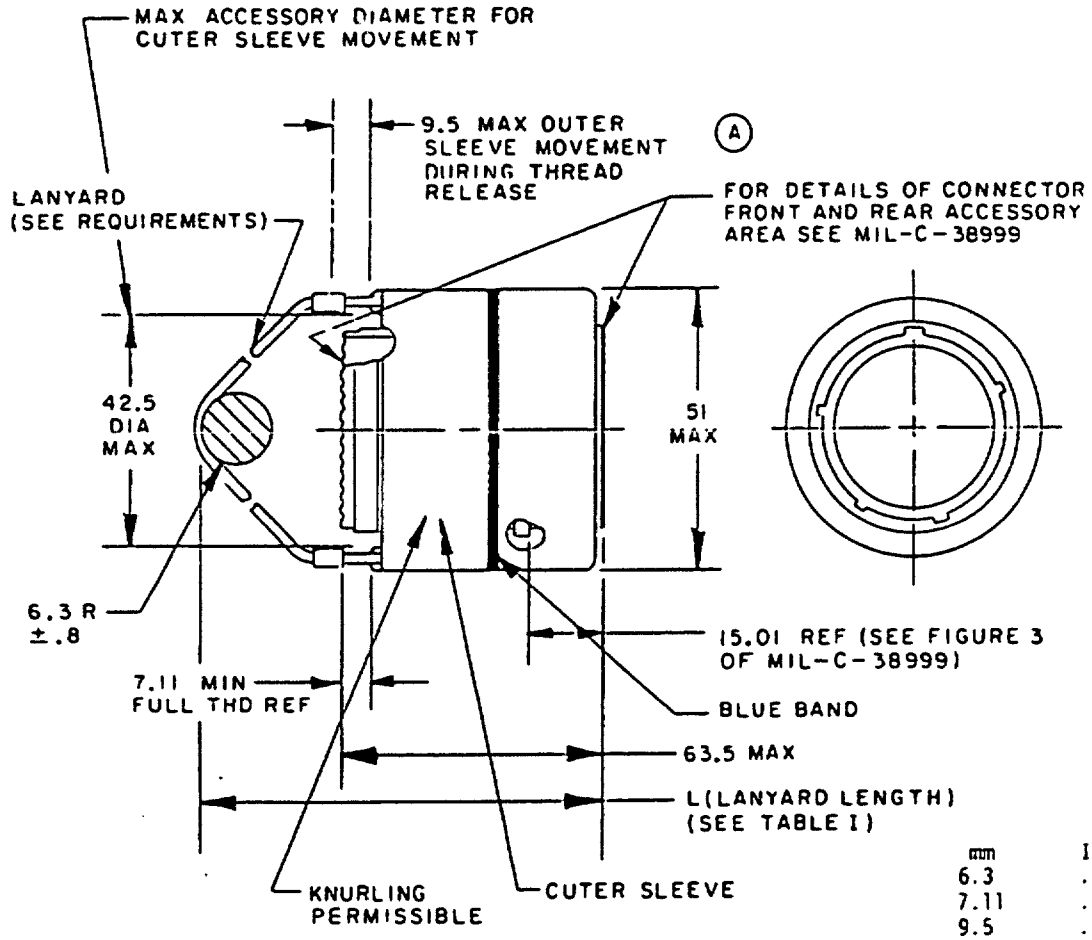
MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, PLUG, LANYARD RELEASE, FAIL-SAFE, REMOVABLE CRIMP CONTACTS, PINS, SERIES III, SHELL SIZE 25, METRIC FOR MIL-STD-1760 APPLICATIONS

Ⓐ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connector described herein shall consist of this specification and the latest issue of MIL-C-38999.



mm	Inch
6.3	.25
7.11	.280
9.5	.37
15.01	.591
42.5	1.67
51	2.0
63.5	2.50

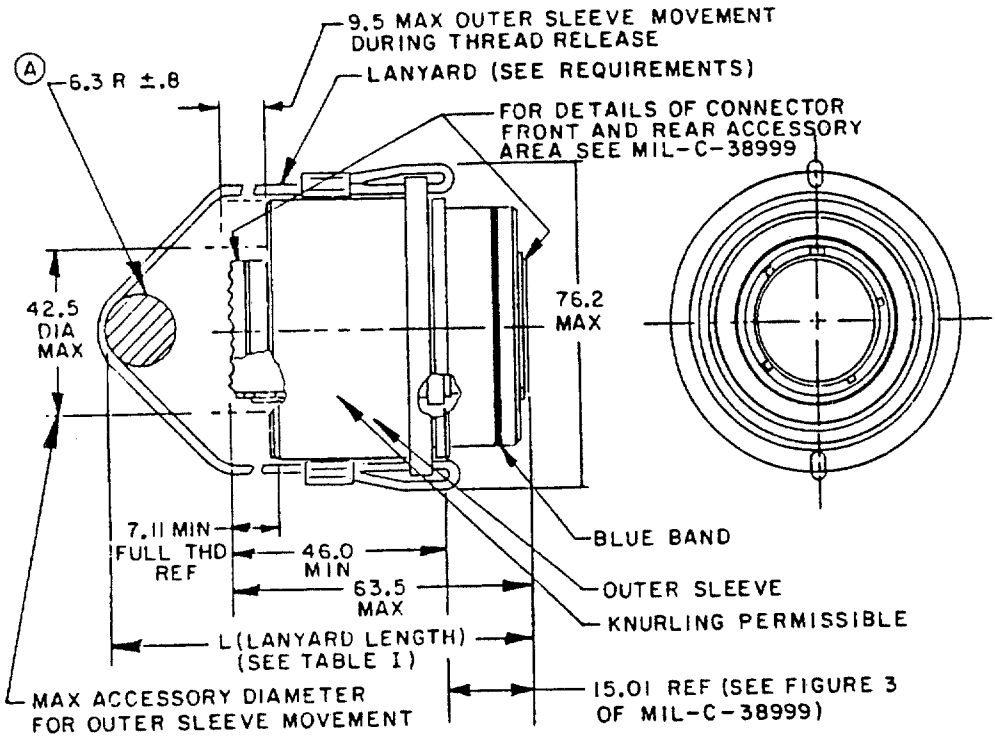
NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are given for general information only.
3. Polarizing position "I" is shown.
4. EMI grounding feature required on this connector.
5. The government may stock, store, and issue this configuration.

FIGURE 1. Plug, class W, type 1.

AMSC N/A
FSC 5935

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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mm	Inch
0.8	.03
6.3	.25
7.11	.280
9.5	.37
15.01	.591
42.5	1.67
46.0	1.81
63.5	2.50
76.2	3.00

NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are given for general information only.
3. Polarizing position N is shown.
4. EMI grounding feature required on this connector.
5. The government will not stock, store,and issue this configuration.

FIGURE 2. Plug, class W, type 2.

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REQUIREMENTS:

Dimensions and configurations: See figure 1. Interface dimensions shall conform to MIL-C-38999.

This connector mates with and is designed for use with receptacle connectors DOD-C-38999/20 and /24.

Ⓐ For insert arrangements: Only 25-11 and 25-20 of MIL-STD-1560.

Lanyard:

a. 1.57 mm minimum diameter, seven strands of stainless steel capable of withstanding 890 newtons pull test after assembly with the connector.

b. Coupling design optional. Cable shall be covered with a suitable protective sleeving to preclude possible chaffing of wires.

Connector shall disengage from any coupling condition including partially mated.

Connector design shall incorporate a swivel action for the lanyard to prevent twisting of the cable.

Ⓐ Spare contacts: The twin-axial and coaxial contacts are exempted from the spare contact requirements of MIL-C-38999.

QUALIFICATION:

Connectors shall meet the qualification requirements of MIL-C-38999 with the following exceptions and additions:

Table XI, qualification and group C periodic tests (see MIL-C-38999)

Group 11: Replace coupling torque requirement with pull-separation force.

Durability: Wired connectors shall meet the durability requirements of MIL-C-38999, with the following exception:

The total number of cycles of mating and unmating shall be 500, in the following sequence: 200 cycles of normal mating and unmating, 50 cycles of normal mating with pull-separation unmating, 200 cycles of normal mating and unmating, and 50 cycles of normal mating with pull-separation unmating. The lanyard release velocity during the pull-separation unmating cycles shall be 9.15 m/s.

Pull-separation force: In addition to mating and unmating by normal coupling ring rotation, the connector shall be capable of lanyard-pull separation at any angle within 15° of the normal axis. Each connector shall have one straight pull and one pull at 15° from straight, with a pull rate not exceeding 13 cm/s. The test will be at -65°C, +0°C, -5°C at ambient, and at the maximum temperature of the specified class. The test will be conducted within 3 minutes after removal from the temperature chamber without forced heating or cooling. Maximum separation forces shall be 400 newtons for a straight pull and 445 newtons for a 15° pull.

Fail-safe disengagement: Connectors shall be partially mated with the plug coupling ring rotated 50 percent of full coupling. Pull-separation shall be accomplished within the limits as specified above.

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Vibration: Wired mated connectors shall meet the vibration requirements of MIL-C-38999 with the following exceptions:

a. Sine vibration: Connectors shall be subjected to the test specified in method 204, test condition G, of MIL-STD-202.

b. Random vibration: Connectors shall be subjected to the test specified in method 2005 of MIL-STD-1344, test condition VI, letter J, ambient temperature. Duration shall be 8 hours in the longitudinal direction and in a perpendicular direction, for a total of 16 hours.

c. The QPL evaluating activity will define the accessory load and cable to be used in the random and sine vibration tests.

Ⓐ Ice resistance: The mated, wired connectors with accessories attached shall be placed in a chamber and the temperature reduced and stabilized such that the item is maintained at -18°C, +0°C, -5°C (0°F) for 1 hour. After stabilization of the chamber temperature, the test item shall be sprayed with water precooled to 2°C +5°C, -0°C (35°F), for a period of 5 minutes. The test item shall be located a maximum of 305 mm (12 inches) from the spray nozzle. The entire test item shall be exposed to the spray. After completion of water spray, the test item shall remain in the chamber -18°C, +0°C, -5°C (0°F) for an additional 30 minutes. Upon completion of the 30-minute cold soak period, the test item shall be removed from the chamber and immediately (within 2 minutes) subjected to uncoupling by use of the lanyard mechanism. The force required to separate the connector shall not exceed 400 newtons straight pull and 445 newtons 15° pull by more than 50 percent.

Dust (fine sand): Upon completion of the dust test, the test items shall be removed from the chamber and immediately subjected to uncoupling by use of the lanyard mechanism. The force required to separate the connectors shall not exceed the required values specified for straight pull separation by more than 25 percent.

Part number example: D38999/ 31 W E 20 N 1

Ⓐ For figure 1: 1/

Specification number prefix: _____

Specification sheet number: _____

Class: _____

Lanyard length code: 2/

Ⓐ Insert arrangement: 3/ _____

Polarizing positon: _____

Ⓐ Type number (see figure 1): _____

Ⓐ For figure 2: D36999/ 31 W E 20 N 2

Specification number prefix: _____

Specification sheet number: _____

Class: _____

Lanyard length code: 2/ _____

Insert arrangement: 3/ _____

Polarizing positon: _____

Type number (see figure 2): _____

1/ The Government may stock, store, and issue type 1 only (see figure 1).
2/ See table I.
3/ Only 25-11 and 25-20 of MIL-STD-1560 apply.

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TABLE I. Lanyard length code.

Code	L ± 6
E	153 (6.0)
F	166 (6.5)
G	178 (7.0)
H	191 (7.5)

QPL evaluating activity: DESC-EQ

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, MI
Navy - EC, MC, OS
Air Force - 11, 15, 99
DLA - ES

User activity:
Air Force - 80

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5935-3585)

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